# Software Requirements Specifications

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Github repo : https://github.com/DanteArias7/COSC-310-Group-Lucky-13
Project backlog link : https://github.com/users/DanteArias7/projects/6

# Software Requirements Specification Document

## Project Description and Goal:

Group Lucky 13 will be creating a full-stack dockerized multiuser system which uses modern software engineering practices in order to demonstrate proper system architecture, modularity and testing. The project will involve developing a backend system for a food delivery service using a provided dataset. The primary focus of the project is the backend functionality which includes: data ingestion, business logic implementation, and API development. The group will use the existing dataset to model entities such as users, restaurants, drivers, and orders. The created system will support common features such as: searching and filtering data, managing orders, calculating prices, and creating order workflows.

The goal of this project is having hands-on experience with real-world backend tasks, as well as learning data validation, consistency management, and scalable system design. Group Lucky 13 aims to master back-end system design while following best software engineering practices.

# Functional Requirements:

**Feature 1:** The system will allow users and restaurant owners/managers to create accounts and log in. It will handle user authentication (login), authorization (what each user is allowed to do), and basic user identity management. Different roles such as regular users and restaurant owners/managers will be supported.

a. ***System & Software Requirements:***
   i. The system shall allow customers to create an account with the necessary information including name, address, and phone number.
   ii. The system shall allow restaurant owners/managers to set up a profile for their establishment.
   iii. The system shall allow users to log in using identifying information of an email or phone number combined with a password.
   iv. The system shall restrict customers or other restaurant owners from being able to edit a restaurant's profile.

b. ***User stories:***
   i. As a customer, I would like to be able to save the necessary details to complete an order, so I do not have to reenter them every time I order.

      Acceptance Criteria: A customer enters necessary information into their profile like name, address, and phone number when creating their account, these details are saved as default, when creating a new order they are automatically used.

   ii. As a customer or restaurant owner/manager I would like to be able to edit my profile information, so I can update information about my profile if it changes.

      Acceptance Criteria: A restaurant owner/manager clicks on their profile, clicks the edit profile button, is able to change relevant information, clicks save changes, changes are reflected in their profile.

**Feature 2:** The system will store information about restaurants and their menus. It will ensure that data is valid, properly connected (for example, menu items must belong to a restaurant), and that basic constraints are enforced, such as preventing invalid or missing values.

a. ***System & Software Requirements:***
   i. The system shall allow restaurant owners/managers to display their restaurant's information such as menus, hours, prices, and phone number.
   ii. The system shall allow restaurant owners/managers to change/update identifying information about their restaurant.
   iii. The system must ensure all menu items are complete, meaning they have name, description and price.
   iv. The system must ensure that every restaurant has a menu.

b. ***User stories:***
   i. As a restaurant owner/manager, I would like to be able to indicate if a menu item is sold out, so that I do not have customers ordering items I do not have.
      Acceptance Criteria: Restaurant owner/manager opens menu, clicks edit menu button, selects sold out option on necessary menu items, item now displays as sold out, a notification displays if the change is successful or not.

   ii. As a restaurant owner/manager I would like to be able to add, update, or remove menu items, so that I can keep my menu up to date.
      Acceptance Criteria: Restaurant owner/manager opens menu, clicks edit menu button, makes necessary changes to menu, clicks save changes button, menu changes are reflected, a notification is sent indicating if the changes were successful or not.

   iii. As a restaurant owner/manager, I would like to be able to put discounts on certain items, so I can offer deals to my customers.
      Acceptance Criteria: Restaurant owner/manager opens menu, clicks edit menu button, clicks add discount button, adds and saves discount, discount is reflected on menu for customers.

**Feature 3:** The system will allow users to browse restaurant menus and search for items or restaurants. Backend logic will handle filtering, searching, and returning paginated results.

a. ***System & Software Requirements:***
   i. The system shall allow customers to search for menu items or restaurants by name.
   ii. The system shall allow customers to filter by things like cuisine type, location, price range, or dietary options (gluten free, vegan, etc).
   iii. The system should show only menu items which are available, from stores which are open.

b. ***User Stories:***
   i. As a customer, I want to be able to search for a restaurant directly, so that I can quickly access the menu I want to order from.

      Acceptance criteria: Customer inputs search terms into search bar, new paginated results are returned with only restaurants or menu items that match the search terms.

   ii. As a customer, I want to be able to see a list of only restaurants that accommodate my dietary restrictions, so that I can quickly find the dishes which I'm able to order.

      Acceptance criteria: Each restaurant's menu items have tags which indicate cuisine, dietary restrictions, etc, customers can select which tags to filter by, results return only restaurants which have the corresponding tags.

   iii. As a customer, I want to see only the restaurants which can deliver food at the current or scheduled time, so that I can see what's actually available.

      Acceptance criteria: When no search terms are entered or filters are applied, the full restaurant list should only show restaurants which are currently open, and each restaurant menu should only display the dishes which are currently available.

**Feature 4:** The system will allow users to create and manage food orders. It will ensure that orders are consistent, correctly stored, and follow business logic for the domain (for example, an order cannot be modified after it is completed).

a. ***System & Software requirements:***
   i. The system shall allow customers to add multiple items from the same restaurant, but prevent items from multiple restaurants being added to the same cart.
   ii. The system shall not allow changes from the customer after the order has been submitted and accepted by the restaurant and delivery driver.
   iii. The system should allow restaurant owners/managers and drivers to reject an order.
   iv. The system should automatically reject orders which are invalid (restaurant is closed, item is out of stock, etc).
   v. The system should not allow changes from any user (restaurant, delivery driver, or customer) after the order has been fulfilled (prepared and delivered).

b. ***User Stories:***
   i. As a restaurant owner, I want to be able to reject orders, so that in the case where we are not able to fulfill a request the order will not go through.
      Acceptance criteria: A popup comes up with order details, accept/reject button at the bottom.

   ii. As a customer, I want to be able to review my past orders, so that I can access those details if needed
      Acceptance criteria: Separate section with all past orders in chronological order, click on one to view more details

**Feature 5:** The system will manage delivery-related information. It will support assigning deliveries and tracking basic delivery status as part of the backend logic.

a. ***System & Software Requirements***
   i. The system shall prevent the same delivery being assigned to multiple drivers simultaneously.
   ii. The system shall allow reassignment of deliveries that have not yet been picked up.
   iii. The system shall calculate and display estimated delivery time based on distance and current traffic conditions.

b. ***User Stories:***
   i. As a delivery driver, I want to view available delivery assignments, so that I can choose which deliveries to accept.

Acceptance Criteria: A list showing the restaurant, customer address, order value, and delivery distance.

ii. As a delivery driver, I want to update delivery status, so that customers can track their order.

Acceptance Criteria: Status updates are reflected in real time. The customer gets a notification when status is updated.

iii. As an admin, I want to assign deliveries to available drivers, so that orders are delivered efficiently.

Acceptance Criteria: Only assign deliveries to active drivers. Consider driver location when assigning deliveries. Make sure deliveries are not double-assigned.

iv. As a customer, I want to view my delivery status and estimated delivery time, so that I know when to expect my food.

Acceptance Criteria: Real-time status is visible. ETA calculated based on distance and current status.

**Feature 6:** The system will calculate the total cost of an order, including item prices, delivery fees, and taxes. These calculations will follow predefined business rules implemented in the backend.

***a. System & Software Requirements:***

i. The system shall calculate the order subtotal as the sum of the price of all items in the cart + delivery fee + tax - discount.

ii. The system shall calculate delivery fee based on the distance between the restaurant and the customer address.

iii. The system shall display itemized price breakdown to the customer before checkout.

***b. User Stories:***

i. As a customer, I want to see the total cost breakdown, so that I understand what I am paying for.

Acceptance Criteria: Itemized breakdown provided. Delivery fee based on distance. Total price displayed clearly.

ii. As a restaurant owner/manager, I want delivery fees to be calculated based on distance so that pricing is fair.

Acceptance Criteria: A minimum base fee of $1.00, plus an additional $0.15 charge per kilometer.

**Feature 7:** The system will simulate payment processing. No real payment gateway will be used, but the system will follow the correct workflow for accepting or rejecting a payment and updating the order status.

***a. System & Software Requirements:***

i. The system shall allow the customer to pay for an order.
ii. The system shall simulate payment processing without using a real payment gateway.
iii. The system shall determine whether a payment is accepted or rejected based on pre-defined rules.
iv. The system shall update the order status based on the payment outcome.
v. The system shall notify the customer of the payment result.

***b. User Stories:***

i. As a customer, I want to pay for my order, so that I can complete my purchase.
Acceptance criteria: The order exists, the order is in a payable state (e.g., Pending Payment), payment is completed.

ii. As a customer, I want to know whether my payment was accepted or rejected so that I know if my order was successful.
Acceptance criteria : Payment for an order is completed, a clear success or failure message is shown, the order status is updated accordingly.

iii. As a customer, I want to retry payment after a failure, so that I can still complete my order.
Acceptance criteria: Payment fails, customer is notified, retry button is pressed, order is reprocessed.

iv. As a restaurant owner/manager, I want to see only paid orders, so that I prepare food only for confirmed purchases.
Acceptance criteria: Orders appear only after payment is accepted, failed payments do not trigger order preparation.

v. As an admin, I want to view failed payment attempts, so that I can monitor system behavior and issues.
Acceptance criteria : Failed payments are logged, failure reasons are visible.

**Feature 8:** The system will generate notifications or events when important actions occur, such as order creation or status changes.

a. ***System & Software Requirements:***
   i. The system shall generate notifications whenever important actions occur, such as order creation, payment outcome, or order status changes.
   ii. The system shall ensure that relevant notifications are visible to customers (e.g. order creation or order status changes).
   iii. The system shall ensure that relevant notifications are visible to the restaurant owners/managers (e.g. new paid orders).

b. ***User stories:***
   i. As a customer, I want to see a confirmation notification when my order is created, so that I know the system has recorded my order.
      Acceptance criteria: The order is successfully created in the system, the notification includes the order number, the notification is only visible to the customer who created the order.

   ii. As a restaurant owner/manager, I want to be notified when a new paid order is received, so that I know when to begin preparing food.
      Acceptance criteria: Only orders with status Paid are visible to the restaurant manager, a notification appears when a payment is accepted.

   iii. As a customer, I want to see updates when my order status changes, so that I can track my order.
      Acceptance criteria: A notification is generated whenever the order status changes, the notification clearly states the new order status along with the order number, the displayed status matches the backend order state.

   iv. As a customer , I want to receive a notification whenever an error occurs (item is out of order, payment failure, restaurant closed) so that I clearly understand what went wrong.
      Acceptance criteria: A notification is displayed whenever an error happens, the notification explains the error in clear , user-friendly language with no technical terms.

# Non- Functional Requirements:

**Feature 1:**

a. ***Performance:***
   i. The system should be able to create an account within 5 seconds once the information has been added
b. ***Reliability:***
   i. The system should properly authenticate the user 99% of the time.
c. ***Security:***
   i. The system should hide private data of each user, only displaying the necessary info to complete an order.
d. ***Usability:***
   i. The user should be able to create an account within 3 minutes.

**Feature 2:**

a. ***Performance:***
   i. The system should display menu changes within 1 second.
b. ***Reliability:***
   i. The system should properly reflect menu changes or display an error if unsuccessful.
   ii. The system should display a confirmation message when menu changes are saved.
c. ***Security:***
   i. The system should hide private data of each user, only displaying the necessary info to complete an order.
   ii. Customers should not be able to change restaurants' menus.
d. ***Usability:***
   i. The system should be able to start a new order

**Feature 3:**

a. ***Performance:***
   i. The results page should load in under 5 seconds
b. ***Reliability:***
   i. The system should successfully return search results for 99% of user queries without failure or timeout
c. ***Security:***
   i. Only publicly available info about the restaurant (such as the menu items with prices, restaurant hours, etc) should be shown on the customer facing side.
d. ***Usability:***
   i. Customers should be able to filter by cuisine type, location, price range, or dietary options.

**Feature 4:**

a. ***Performance:***
   i. Updates and order requests should be reflected in the system in under 3 seconds
b. ***Reliability:***
   i. Changes to an order should be persistent in the system for all users within 2 seconds of submission, 95% of the time
   ii. The system should process 99% of order requests without failure or timeout
c. ***Security:***
   i. The system should not allow editing of orders from the customer side, after the order has been submitted and accepted
d. ***Usability:***
   i. The system should only show actions which the user is able to take

**Feature 5:**

a. ***Performance:***
   i. The system shall retrieve the list of all available drivers within 1 second for the driver assignment interface.
   ii. The system should load delivery detail pages within 2 seconds.
b. ***Reliability:***
   i. The system shall prevent data loss during loss during delivery status updates using database transactions.
   ii. The system shall retry failed notifications up to 3 times before logging as failed.
c. ***Usability:***
   i. The system shall display delivery status updates to customers in real time.
d. ***Security:***
   i. The system will only allow authenticated drivers to update the delivery status.
   ii. The system will prevent drivers from viewing deliveries assigned to other drivers.

**Feature 6:**

a. ***Performance:***
   i. The system shall re-calculate prices within 1 second when the cart is modified.
b. ***Reliability:***
   i. The system shall prevent negative prices from being calculated or stored.
c. ***Usability:***

i. The system shall display a clear price breakdown to the customer right before checkout.
ii. The system will display an error message for things such as: minimum order price, distance error, and cart expiration error.

d. ***Security:***
i. The system will make sure price calculation results are only visible to the requesting user.

**Feature 7:**

a. ***Performance:***
i. The system shall complete simulated payment processing and return a result within 1 second for at least 95% of payment requests.
ii. The system shall update the order status within 1 second after a simulated payment decision is made.

b. ***Reliability:***
i. For each payment attempt, the system shall generate exactly one payment outcome (accepted or rejected).
ii. The system must not mark a payment as complete unless the order is also updated correctly.

c. ***Security:***
i. The system shall not store or process real credit card numbers or bank details.
ii. Only authenticated customers shall be allowed to initiate simulated payment requests.

d. ***Usability:***
i. The system shall provide a clear success or failure message to the customer immediately after payment processing completes

**Feature 8:**

*a.* ***Performance:***
*i.* The notifications shall be visible within 1 second after the corresponding action happens.

b. ***Reliability:***
*i.* The notifications shall accurately reflect the current state of the associated order at the time they are generated.

c. ***Security:***
i. Notifications shall be visible only to users with appropriate roles (e.g. customers see their own order notifications , restaurant managers see new paid orders notifications)

d. ***Usability:***
i. Notifications shall use clear and understandable language describing the action or status change.

## Domain Requirements:

**Feature 1:**

i. Each restaurant manager/manager should be able to have multiple locations on a single account.

**Feature 2:**

i. The system must allow for limited time specialty menus to be displayed for events.

**Feature 3:**

i. Each restaurant may not have multiple locations (chains will have separate pages for each location)
ii. Each menu item may only have one set price at a time

**Feature 4:**

i. The system must only create an order request after delivery address, cart items, and payment methods have been verified
ii. The system should show the same order info on both the client and restaurant side

**Feature 5:**

i. Each delivery shall be linked to exactly one order.
ii. A driver can only be assigned to one driver at any given time.
iii. Driver availability states are: available, busy, and offline.

**Feature 6:**

i. A price breakdown of the order shall show itemized costs visible to customers before checkout.
ii. Item price shall be captured when the item was added to cart.
iii. The subtotal of the order must be a positive value greater than zero.

**Feature 7:**

i. An order must only be considered paid after a payment attempt has been successfully accepted.
ii. An order must not be sent to the restaurant unless its payment status is Paid.
iii. Each order may have multiple payment attempts, but at most one successful payment.
iv. Payment processing should always update the order status correctly based on whether the payment succeeds or fails.

**Feature 8:**

i. The notifications must reflect only relevant changes for the receiving role (customer / restaurant manager)

## Contribution:

**Dante Arias:** Feature 1 & 2
**Zheng Xue:** Feature 3 & 4
**Gurkirn Kaur:** Feature 5 & 6
**Varun Garg:** Feature 7 & 8

Each member created the requirements, added to the traceability matrix, and added to the backlog all issues related to their 2 features. Along with that, we went over each feature's requirements, matrix placement, and backlog issue together as a group ensuring there were no conflicts, and everyone agreed on each point.

# Requirements Traceability matrix

## 2. Requirements traceability matrix

| Core Feature ID | Functional Requirement ID(s) | User Story ID(s) |
|---|---|---|
| **Feature 1** | | |
| **Feat1:** The system will allow users and restaurant owners/managers to create accounts and log in. It will handle user authentication (login), authorization (what each user is allowed to do), and basic user identity management. Different roles such as regular users and restaurant owners/managers will be supported. | **Feat1-FR1:** The system shall allow customers to create an account with the necessary information including name, address, and phone number. | **Feat1-US1:** As a customer, I would like to be able to save the necessary details to complete an order, so I do not have to reenter them every time I order. |
| | **Feat1-FR2:** The system shall allow restaurant owners/managers to set up a profile for their establishment. | **Feat1-US2**: As a customer or restaurant owner/manager I would like to be able to edit my profile information, so I can update information about my profile if it changes. |
| | **Feat1-FR3:** The system shall allow users to log in using identifying information of an email or phone number combined with a password. | |
| | Feat1-FR4: The system shall restrict customers or other restaurant owners from being able to edit a restaurant's profile. | |

| Feature 2 | | |
|---|---|---|
| **Feat2**: The system will store information about restaurants and their menus. It will ensure that data is valid, properly connected (for example, menu items must belong to a restaurant), and that basic constraints are enforced, such as preventing invalid or missing values. | **Feat2-FR1:** The system shall allow restaurant owners/managers to display their restaurant's information such as menus, hours, prices, and phone number. | **Feat2-US1:** As a restaurant owner/manager, I would like to be able to indicate if a menu item is sold out, so that I do not have customers ordering items I do not have. |
| | **Feat2-FR2:** The system shall allow restaurant owners/managers to change/update identifying information about their restaurant. | **Feat2-US2:** As a restaurant owner/manager I would like to be able to add, update, or remove menu items, so that I can keep my menu up to date. |
| | **Feat2-FR3:** The system must ensure all menu items are complete, meaning they have name, description and price. | **Feat2-US3**: As a restaurant owner/manager, I would like to be able to put discounts on certain items, so I can offer deals to my customers. |
| | Feat2-FR4: The system must ensure that every restaurant has a menu. | |
| **Feature 3** | | |
| **Feat3:** The system will allow users to browse restaurant menus and search for items or restaurants. Backend logic will handle filtering, searching, and returning paginated results. | **Feat3-FR1:** The system shall allow customers to search for menu items or restaurants by name. | **Feat3-US1:** As a customer, I want to be able to search for a restaurant directly, so that I can quickly access the menu I want to order from. |
| | **Feat3-FR2:** The system shall allow customers to filter by cuisine type, location, price range, or dietary options (gluten free, vegan, etc). | **Feat3-US2:** As a customer, I want to be able to see a list of only restaurants that accommodate my dietary restrictions, so that I can quickly find the dishes which I'm able to order. |

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| | **Feat3-FR3:** The system should show only menu items which are available, from stores which are open | **Feat3-US3:** As a customer, I want to see only the restaurants which can deliver food at the current or scheduled time, so that I can see what's actually available. |
| **Feature 4** | | |
| **Feat4:** The system will allow users to create and manage food orders. It will ensure that orders are consistent, correctly stored, and follow business logic for the domain (for example, an order cannot be modified after it is completed). | **Feat4-FR1:** The system shall allow customers to add multiple items from the same restaurant, but prevent items from multiple restaurants being added to the same cart | **Feat4-US1:** As a customer, I want to be able to review my past orders, so that I can access those details if needed |
| | **Feat4-FR2:** The system shall not allow changes from the customer after the order has been submitted and accepted by the restaurant and delivery driver | **Feat4-US2:** As a restaurant owner/manager, I want to be able to reject orders, so that in the case where we are not able to fulfill a request the order will not go through. |
| | **Feat4-FR3:** The system should allow restaurant owners and drivers to reject an order | |
| | **Feat4-FR4:** The system should automatically reject orders which are invalid (restaurant is closed, item is out of stock, etc) | |

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|---|---|---|
| | **Feat4-FR5:** The system should not allow changes from any user (restaurant, delivery driver, or customer) after the order has been fulfilled (prepared and delivered) | |
| **Feature 5** | | |
| **Feat5:** The system will manage delivery-related information. It will support assigning deliveries and tracking basic delivery status as part of the backend logic. | **Feat5-FR1:** The system shall prevent the same delivery being assigned to multiple drivers simultaneously. | **Feat5-US1:** As a delivery driver, I want to view available delivery assignments, so that I can choose which deliveries to accept. |
| | **Feat5-FR2:** The system shall allow reassignment of deliveries that have not yet been picked up. | **Feat5-US2:** As a delivery driver, I want to update delivery status, so that customers can track their order. |
| | | **Feat5-US3:** As an admin, I want to assign deliveries to available drivers, so that orders are delivered efficiently. |
| | Feat5-FR3: The system shall calculate and display estimated delivery time based on distance and current traffic conditions. | **Feat5-US4:** As a customer, I want to view my delivery status and estimated delivery time, so that I know when to expect my food. |
| **Feature 6** | | |
| **Feat6:** The system will calculate the total cost of an order, including item prices, delivery fees, and taxes. These calculations will follow predefined business rules implemented in the | **Feat6-FR1:** The system shall calculate the order subtotal as the sum of the price of all items in the cart + delivery fee + tax - discount. | **Feat6-US1:** As a customer, I want to see the total cost breakdown, so that I understand what I am paying for. |

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|---|---|---|
| backend. | **Feat6-FR2:** The system shall calculate delivery fee based on the distance between the restaurant and the customer address. | |
| | | **Feat6-US2**: As a restaurant owner, I want delivery fees to be calculated based on distance so that pricing is fair. |
| | **Feat6-FR3:** The system shall display itemized price breakdown to the customer before checkout. | |
| **Feature 7** | | |
| **Feat7:** The system will simulate payment processing. No real payment gateway will be used, but the system will follow the correct workflow for accepting or rejecting a payment and updating the order status | **Feat7-FR1:** The system shall allow the customer to pay for an order. | **Feat7-US1:** As a customer, I want to pay for my order, so that I can complete my purchase. |
| | **Feat7-FR2:** The system shall simulate payment processing without using a real payment gateway. | **Feat7-US2:** As a customer, I want to know whether my payment was accepted or rejected so that I know if my order was successful. |
| | **Feat7-FR3:** The system shall determine whether a payment is accepted or rejected based on pre-defined rules. | **Feat7-US3:**As a customer, I want to retry payment after a failure, so that I can still complete my order. |
| | **Feat7-FR4:** The system shall update the order status based on the payment outcome. | **Feat7-US4:** As a restaurant manager, I want to see only paid orders, so that I prepare food only for confirmed purchases. |
| | **Feat7-FR5:** The system shall notify the customer of the payment result. | **Feat7-US5:** As an admin, I want to view failed payment attempts, so that I can monitor system behavior and issues. |

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|---|---|---|
| **Feature 8** | | |
| **Feat8:** The system will generate notifications or events when important actions occur, such as order creation or status changes. | **Feat8-FR1:** The system shall generate notifications whenever important actions occur, such as order creation, payment outcome, or order status changes. | **Feat8-US1:** As a customer, I want to see a confirmation notification when my order is created, so that I know the system has recorded my order. |
| | **Feat8-FR2:** The system shall ensure that relevant notifications are visible to customers (e.g. order creation or order status changes). | **Feat8-US2:** As a restaurant manager, I want to be notified when a new paid order is received, so that I know when to begin preparing food. |
| | | **Feat8-US3:** As a customer, I want to see updates when my order status changes, so that I can track my order. |
| | **Feat8-FR3**: The system shall ensure that relevant notifications are visible to the restaurant managers (e.g. new paid orders). | **Feat8-US4:** As a customer , I want to receive a notification whenever an error occurs (item is out of order, payment failure, restaurant closed) so that I clearly understand what went wrong. |